“貝靈”凝血纖維蛋白原注射液　　　　衛署菌疫輸字第 000439 號

Haemocomplettan®P　　　　本藥限由醫師使用

使用藥品前，請先仔細閱讀此說明書的所有資訊，因其可能包含給您的重要資訊。
- 請妥善保管說明書，以便日後若有需要時參考。
- 若有任何其他的疑慮，請詢問您的醫師或藥師。
- 此藥品處方只開立給您，請勿給他人使用，可能會對他人造成傷害，即使他們的症狀與您的相同。
- 若您有任何副作用發生，請告知您的醫生或藥師。包含任何說明書上沒有載明的可能副作用。(請參見章節 4.8)

1. 產品名稱

“貝靈”凝血纖維蛋白原注射液 1g
注射/輸注用溶液之粉末

2. 定性與定量的組成

Haemocomplettan 為供注射或靜脈輸注用粉末，每小瓶含 1 公克之人體纖維蛋白原(Human fibrinogen)。

Haemocomplettan 1g 可與 50ml 注射用水配製，配置後為每毫升含 20 毫克人體纖維蛋白原(20mg/mL)。

可凝固之纖維蛋白原(clottable fibrinogen)的含量是依據歐洲藥典人體纖維蛋白原章節所訂定。
經認定具已知功效之賦形劑：每 1g 纖維蛋白原可達 164mg (7.1mmol)的鈉。
詳細之賦形劑清單，請參見章節 6.1。

3. 劑型

注射/輸注用溶液之粉末。
白色粉末

4. 臨床特性

4.1 適應症

出血症

說明：
1.先天性纖維蛋白原過低、異常或無纖維蛋白原血症
2.由於以下症狀造成之後天性低纖維蛋白原血症
　a.嚴重肝實質傷害所致之合成失調
　b.播散性血管內凝血症（DIC）與過度纖維蛋白溶解反應（Hyperfibrinolysis）所造成之過度耗損。

臨床上最易引起纖維蛋白原缺乏的症狀如下：
產科併發症、輸血錯誤及中毒後引起之溶血、各種型態之休克、外傷、手術及肺臟、胰臟、子宮與前列腺之腫瘤、肝硬化、急性白血病（血癌）。

4.2 使用劑量及投與方式

必須在有治療凝血失調經驗的醫師的監督下，開始進行治療。

使用劑量

替代治療的劑量與使用期間，必須視疾病的嚴重性、出血的部位及程度，與患者的臨床狀況而定。

為計算病患投與劑量，應測定纖維蛋白原（具功能性的）之濃度，個別患者投與的量與頻率也應依據常規測量的血漿纖維蛋白原濃度來判斷，並持續監測患者的臨床狀況和其他替代療法的使用。

正常血漿纖維蛋白原濃度範圍為 1.5-4.5 g/l。關鍵血漿纖維蛋白原濃度(critical plasma fibrinogen level)約 0.5-1.0g/l 以下時，即可能發生出血。因此在重大外科手術時，藉由凝血試驗精確地監測替代療法是必要的。

1. <u>先天性低(hypo-)、異常(dys-)或無纖維蛋白原血症(afibrinogenemia)及已知出血傾向的患者之預防。</u>

為預防在手術過程中發生的大量出血現象，建議使用預防性治療，提升纖維蛋白原濃度至 1 g/l 並且維持纖維蛋白原在此濃度至安全止血，及 0.5g/l 以上直到傷口癒合完全。

於手術過程或針對出血事件的治療，需依照以下公式計算劑量：

$$\underset{\text{(mg/kg 體重)}}{\text{纖維蛋白原劑量}} = \frac{[\text{目標濃度(g/l)} - \text{測量濃度(g/l)}]}{0.017\ \text{(g/l 每 mg/ kg 體重)}}$$

合適的後續劑量（注射劑量及頻率）應以患者的臨床狀況和實驗室報告為依

據。

纖維蛋白原之生物半衰期為 3-4 天。因此當缺乏消耗量時，通常不需要重複投與人體纖維蛋白原。累積現象會發生於預防性使用的重複投與，劑量及注射頻率應依據醫生給予患者的治療目標而決定。

2. 出血治療

成人

針對手術前後期間出血，通常投與 1-2g（或 30mg/kg 體重），視需要後續輸注。

嚴重出血的狀況，如產科用(obstetric use) / 胎盤剝離(abruption placenta)，可能需要大量（4-8g）的纖維蛋白原。

孩童

劑量應依體重及其臨床需要決定，但大多為 20-30 mg/kg。

投與方式

靜脈輸注或注射。

Haemocomplettan 應依據章節 6.6 配製。配製後的溶液應於投與前回溫至室溫或體溫，然後以患者最舒適的速度緩慢注射或輸注。每分鐘的注射或輸注速度約不超過 5ml。

4.3 禁忌症

對於活性成分或任何章節 6.1 所列之賦形劑過敏者。

禁用於血栓或心肌梗塞，除非在非常嚴重可能危及生命之出血情況下。

4.4 特別警語及使用注意事項

當先天性缺乏之患者使用人體纖維蛋白原治療時有其風險，尤其是投與高劑量或重複給藥時。在給予患者纖維蛋白原時，應嚴密的觀察血栓的徵兆及症狀。

有冠狀心臟疾病或心肌梗塞病史的患者、有肝臟疾病的患者、在手術前或後的患者、新生兒或具血栓栓塞或瀰漫性血管內凝血症風險的患者，應衡量使用人體血漿纖維蛋白原之潛在利益及血栓栓塞併發症的風險，同時應執行謹慎及嚴密的監控。

後天性低纖維蛋白原血症與所有凝血因子（不僅是纖維蛋白原）及抑制子的低血漿濃度有關，因此使用含有凝血因子的血液製劑治療時應予以評估。謹慎監控凝血系統是必須的。

若過敏或類過敏性反應發生時，應立即停止注射 / 輸注。發生過敏性休克時，應實施休克標準醫療程序。

使用凝血因子替代療法於其他先天性缺乏時，已觀察到抗體反應，但目前尚無纖維蛋白原之相關數據。

Haemocomplettan **特定賦形劑相關之重要資訊**

Haemocomplettan 每 1g 纖維蛋白原可達 164mg (7.1mmol)的鈉。如果使用建議起始劑量為 70mg/kg 體重，則患者每公斤體重相關之鈉為 11.5mg (0.5mmol)。使用在限鈉飲食患者時應予考慮。

病毒安全性

使用由人體血液或血漿製備成的藥劑時，為了避免傳染症，採取的標準措施包括：

挑選捐贈者、針對個別捐贈及混合血漿做特定指標檢測和使用有效的製造程序去活化 / 去除病毒。除此之外，當投與由人體血液或血漿配製而成的藥品，其傳輸感染媒介的可能性是無法被完全排除的。這也適用於未知或新興的病毒和其他病原體。

這些措施被認定為對具套膜的病毒例如human immunodeficiency (HIV)、hepatitis B virus (HBV)和 hepatitis C virus (HCV)及不具套膜的病毒hepatitis A virus (HAV)有效。

這些措施可能對不具套膜的病毒例如parvovirus B19 效果有限。

Parvovirus B19 的感染症可能會嚴重影響懷孕的婦女（胎兒感染症）以及免疫不全或紅血球增生（eg. 溶血性貧血）的患者。

適當的疫苗接種（A 型肝炎以及 B 型肝炎）應用於常規/重複接受人體纖維蛋白原產品的患者。

強烈建議每次投與Haemocomplettan 給患者時，產品名稱及批號必須要記錄，以維持患者及產品批次之間的聯繫。

4.5 與其他藥物的交互作用及其他形式之交互作用

目前尚未發現人體血漿纖維蛋白原產品與其他藥物之交互作用。

4.6 生殖、懷孕及哺乳

懷孕

尚未施行Haemocomplettan 之動物生殖研究（參見章節 5.3）。由於是人類來源之活性成分，其與患者自身蛋白質的分解代謝是相同的。這些人體血液的生理組成對於生殖或是胎兒未預期造成副作用。

目前尚未建立人體血漿纖維蛋白原產品使用於人體懷孕期間的安全性對照臨床試驗。

纖維蛋白原產品治療產科併發症之臨床經驗，顯示對於懷孕期間或胎兒或新生兒的健康無不良影響是可預期的。

哺乳

Haemocomplettan 是否會分泌於人類乳汁仍未知。目前尚未建立人體血漿纖維蛋白原產品於哺乳期間使用之安全性的臨床對照試驗。
對授乳孩童的風險是無法排除的。應考量哺乳對於孩童的利益以及治療對於女性的利益，以決定是否中斷哺乳或是中斷／避免以Haemocomplettan 治療。

生殖

未有Haemocomplettan 關於影響生殖之相關數據。

4.7 對駕駛及操作機械能力上的影響

Haemocomplettan 對駕駛能力及操作機械無任何影響。

4.8 不良反應

a. 安全特徵摘要
不常觀察到過敏或類過敏性反應。報導與過敏/類過敏性反應相關的事件包含：全身性蕁麻疹、紅疹、呼吸困難、心搏過速、噁心、嘔吐、冷顫、發熱、胸痛、咳嗽、血壓下降及過敏性休克（請參見章節 4.4）。

有關經臨床試驗確立之施用纖維蛋白原濃縮物併發血栓栓塞事件（TEE）的風險（請參見章節 4.4），於下表詳述。

非常常觀察到發熱現象。

b. 不良反應（ADRs）列表
本表結合臨床試驗與上市後經驗確立的不良反應。下表所示頻率係基於跨兩企業贊助之包含或不包含其他外科手術的主動脈手術臨床試驗（BI3023-2002 及 BI3023_3002）匯集分析而得以下常規：非常常見（≥1/10）；常見（≥1/100 至<1/10）；不常見（≥1/1,000 至<1/100）；罕見（≥1/10,000 至<1/1,000）；非常罕見（<1/10,000）或未知（從可得資料無法估計）。

計算出的頻率係基於不考量對照臂(comparator arm)頻率下的粗發生率。應注意列入分析的臨床試驗中，安慰劑組具較高TEEs 發生率。有鑒於這些試驗僅在包含

或不包含其他外科手術的主動脈手術此類狹小族群內進行，其觀察到的不良反應頻率可能無法真實反應出臨床診療所觀察到的頻率，並且在研究條件外的臨床設置下仍屬未知。

國際醫學用語詞典系統器官分類（MedDRA SOC）	不良反應	頻率（包含或不包含其他外科手術的主動脈手術）
一般疾病及投與部位狀況	發熱	非常常見
免疫系統失調	過敏或類過敏性反應	不常見
血管疾病	血栓栓塞事件*	常見**

*個別病例已經死亡

**根據兩臨床試驗（包含或不包含其他外科手術的主動脈手術）的結果，相較安慰劑而言，以纖維蛋白原治療的受試者具較低的血栓栓塞事件匯集發生率（請參見章節*4.8.c*）

c. 特定不良反應（ADRs）敘述

BI3023_2002 係以人體纖維蛋白原濃縮物（FCH）與安慰劑（生理食鹽水）相比，於接受主動脈修復手術的急性出血患者族群內進行的第二期臨床試驗。

BI3023-3002 係以FCH 與安慰劑（生理食鹽水）相比，於複雜的心血管手術中控制出血的第三期臨床試驗。在BI3023_2002 研究中（N=61），TEE 在纖維蛋白原和安慰劑族群中具相似發生率。在 BI3023_3002 研究中（N=152），相對 FCH 族群，TEE 在安慰劑族群中具有較高發生率。

由企業贊助臨床試驗（BI3023-2002 及 BI3023_3002）所列之不良反應匯集發生率

不良反應	FCH (N=107)	安慰劑 (N=106)
發熱	11 (10.4%)	5 (4.7%)
血栓栓塞事件	8 (7.4%)	11 (10.4%)
過敏或類過敏性反應	1 (0.9%)	0

傳輸感染媒介相關之安全性，參見章節 4.4。

疑似不良反應通報

於醫療產品上市後通報疑似不良反應是重要的。其可持續的監控醫療產品的利益／風險。健康照護專業人士被要求需通報任何疑似的不良反應。

4.9 過量

為避免過量，需常規監測治療期間的纖維蛋白原血漿濃度（參見 4.2）。

當過量時，其發生血栓栓塞之併發症的風險會增加。

5. 藥理學特性

5.1 藥效學特性

藥理治療分類：抗出血劑，人體纖維蛋白原。

ATC code：B02B B01

人體纖維蛋白原（凝血因子I）在 thrombin 的存在下可活化的凝血因子XIII (F XIIIa)以及鈣離子，其可形成穩定且具彈性的三維結構纖維蛋白止血凝塊。

投與人體纖維蛋白原可以使血漿纖維蛋白濃度上升，且可暫時改善纖維蛋白原缺乏的患者的凝血缺陷。

Phase II 樞紐研究評估單一劑量的PK（請參見 5.2 藥物動力學特性）且其同時也提供利用替代性指標的最大血凝塊硬度(MCF)的有效性數據及安全性數據。

對每位受試者，MCF 是以 70mg/kg 體重投與單一劑量Haemocomplettan 之前（為基準點）及投與後一小時來做測定。如以血栓彈性分析(thromboelastometry)檢測，發現Haemocomplettan 對於先天性纖維蛋白原缺乏的患者(無纖維蛋白原血症)能有效提升血凝塊硬度。在急性出血事件的止血功效及與MCF 的相關性，已於上市後研究獲得驗證。

5.2 藥物動力學特性

人體血漿纖維蛋白原是人體血漿中的一般組成，表現如同內原性纖維蛋白。
在血漿中，纖維蛋白原的生物半衰期約 3-4 天。Haemocomplettan 降解之表現與內源性纖維蛋白原相似。

Haemocomplettan 是經由靜脈投與且其可在血漿中所測得之濃度與投與劑量成正比。

有一藥物動力學研究評估先天性無纖維蛋白原血症之受試者投與前後的人體纖維蛋白原濃度之單一劑量的藥物動力學。這個前瞻性、開放式、非對照且多中心的研究共有 5 位女性及 10 位男性參與，年齡區間約 8 到 61 歲（2 個兒童、3 個青少年、10 個成人）。劑量中位數為 77.0 mg/kg 體重（範圍：76.6-77.4 mg/kg）。

血液從 15 位受試者中採樣（14 位可測量）檢測於基礎點及輸注完成達十四天後之纖維蛋白原活性。此外，體內回復(in vivo recovery, IVR)增加量，定義為每 mg/kg 體重劑量可以增加纖維蛋白原血漿濃度之最大值，檢測輸注後達四小時之濃度而定。增加IVR 之中位數為 1.7 (範圍：1.30-2.73) mg/dl 每 mg/kg 體重。下表提供了藥物動力學的結果。

纖維蛋白原活性之藥物動力學結果

Parameter (n=14)	**Mean ± SD**	**Median (range)**
$t_{1/2}$ [h]	78.7 ± 18.13	77.1 (55.73-117.26)
C_{max} [g/l]	1.4 ± 0.27	1.3 (1.00-2.10)
AUC for dose of 70 mg/kg [h•mg/ml]	124.3 ± 24.16	126.8 (81.73-156.40)
Extrapolated part of AUC [%]	8.4 ± 1.72	7.8 (6.13-12.14)
Cl [ml/h/kg]	0.59 ± 0.13	0.55 (0.45-0.86)
MRT [h]	92.8 ± 20.11	85.9 (66.14-126.44)
V_{ss} [ml/kg]	52.7 ± 7.48	52.7 (36.22-67.67)
IVR [mg/dl per mg/kg body weight]	1.8 ± 0.35	1.7 (1.30-2.73)

$t_{1/2}$ = 末端排除半衰期 (terminal elimination half-life)
h = 小時 (hour)
C_{max} = 四小時內之最高濃度 (maximum concentration within 4 hours)
AUC = 曲線下面積 (area under the curve)
Cl = 廓清率 (clearance)
MRT = 平均停留時間 (mean residence time)
V_{ss} = 穩定狀態分布量 (volume of distribution at steady state)
SD = 標準差 (standard deviation)
IVR = 體內回收率 (in vivo recovery)

5.3 臨床前安全性資料

基於單劑量毒性和安全藥理學研究，非臨床資料顯示對人體無特殊危害。

由於注射異質的人體蛋白後會產生抗體，重覆給予劑量的臨床前研究（慢性毒性、致癌性以及致突變性）不能合理的應用於傳統的動物模式。

6. 藥劑學特性

6.1 賦形劑清單

Human albumin,
L-arginine hydrochloride,
sodium hydroxide (for pH adjustment),
sodium chloride,
sodium citrate.

6.2 不相容性

除章節 6.6 所提到的部分之外，本藥品不可與其他藥品、稀釋劑或是溶劑混合。建議使用標準的輸注組於室溫下配製溶液以供靜脈注射使用。

6.3 效期

2年。

配製後產品之物化安定性已經證實室溫下可達 8 小時(最高 +25 °C)。從微生物學觀點來看，本產品應在配製後立即使用。若配製後產品未立即投與, 室溫下貯存不應超過 8 小時(最高 +25 °C)。配製後產品不應儲存於冰箱。

6.4 貯存之特別注意事項

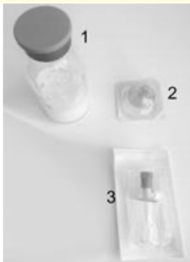
貯存於冰箱中(2℃-8℃)，切勿冷凍！小瓶應放置於外盒內以避光。

6.5 容器材質與內容

100 支以下小瓶裝。

***1g* 包裝** (圖1)

1. 1 小瓶含有 1g 人體纖維蛋白原
2. 過濾器: Pall® Syringe Filter
3. 配液穿刺器: Mini-Spike® Dispensing Pin



(圖 1)

6.6 使用、操作和處置說明

一般說明

-配製和抽取均須在無菌狀態下操作。
-配製後的產品於注射前須以肉眼觀察是否有微粒狀物質或變色。
-溶液應為幾乎無色或淡黃色，清澈略乳白色且 pH 值中性。若溶液混濁或含沉澱物，請勿使用。

配製方法

-將溶劑及粉末之未開封小瓶回溫至室溫或體溫(但不得高於 37ºC)。
-Haemocomplettan 需以注射用水配製(50ml 用於 1g，未附)。
-配製本產品前，請先洗手或穿戴手套。

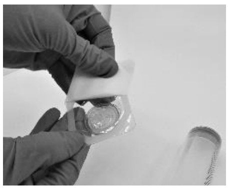
-移去Haemocomplettan 小瓶上之蓋子，露出輸注瓶塞之中心部位。
-以消毒溶液擦拭輸注瓶塞之表面，並待其乾燥。
-以適當的轉注裝置將溶劑注入輸注小瓶中，確保所有粉末均被潤溼。
- 緩緩旋轉小瓶，直到粉末配製完為可供注射的溶液。避免因搖晃過度而形成泡沫。這些粉末最多 15 分鐘內(通常為 5~10 分鐘內)可以配製完成。
-打開Haemocomplettan盒內隨附的配液穿刺器(Mini-Spike® Dispensing Pin)的塑膠包裝（圖2）。


（圖2）

-取出隨附的配液穿刺器後，將其插入含配置後產品的瓶塞上（圖3）。


（圖3）

-插入配液穿刺器後，取下蓋子。 取下蓋子後，請勿觸摸裸露的表面。
-打開Haemocomplettan盒內隨附的過濾器(Pall® Syringe Filter)（圖4）。


（圖4）

-將針筒接到過濾器上（圖5）。


（圖5）

-將已接上濾器的針筒接到配液穿刺器上（圖6）。


（圖6）

−將配置後產品抽至針筒內（圖7）。

（圖 7）

−完成上述步驟後，**請從針筒取下過濾器、配液穿刺器及空瓶。**
適當地丟棄並如往常使用此產品。

-配製後產品應立即使用獨立的注射/輸注管注射(參見章節 6.3)。
-注意不可使血液流入含本產品之注射器中。

任何未使用的產品或廢棄物，必須按照當地衛生法規丟棄。

7. 製造廠：CSL Behring GmbH
地　址：Emil -von-Behring - Str. 76 35041 Marburg, Germany
藥　商：傑特貝林有限公司
地　址：臺北市信義區基隆路 1 段 333 號 16 樓(1612 室)
電　話：(02) 2757-6970

8. 銷售授權碼
-視國家而定-

9. 初次授權日期/重新授權日期
-視國家而定-

10. 最後更新日期：2020 年2 月

Pall® Syringe Filter為Pall Corporation之註冊商標
Mini-Spike® Dispensing Pin為B. Braun Medical Inc之註冊商標

1. NAME OF THE MEDICINAL PRODUCT

Haemocomplettan® P 1g/2g
Powder for solution for injection / infusion – E.P.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Haemocomplettan is presented as a powder for solution for injection or infusion for intravenous administration containing 1g or 2g human fibrinogen per vial.

The product contains 20 mg/ml human fibrinogen after reconstitution with 50 ml water for injections for Haemocomplettan P 1g or 100 ml water for injections for Haemocomplettan P 2g.

The content of clottable fibrinogen is determined according to PH. Eur. monograph for human fibrinogen.

Excipients recognised to have a known effect: Sodium up to 164 mg (7.1 mmol) per 1g fibrinogen.
For a full list of excipients, see section 6.1

3. PHARMACEUTICAL FORM

Powder for solution for injection/infusion.
White powder

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Therapy and prophylaxis of haemorrhagic diatheses in:

- Congenital hypo-, dys- or afibrinogenaemia
- Acquired hypofibrinogenaemia resulting from
  - disorders of synthesis in cases of severe liver parenchyma damage
  - increased intravascular consumption e.g. as a result of disseminated intravascular coagulation, hyperfibrinolysis
  - increased blood loss

The most important clinical pictures associated with a defibrination syndrome are: Obstetrical complications, acute leukaemia especially promyelocytic leukaemia, liver cirrhosis, intoxications, extensive injuries, haemolysis after transfusion errors, operative interventions, infections, sepsis, all forms of shock as well as tumours especially in the lung, pancreas, uterus, and prostate.

4.2 Posology and method of administration

Treatment should be initiated under the supervision of a physician experienced in the treatment of coagulation disorders.

Posology

The dosage and duration of the substitution therapy depend on the severity of the disorder, location and extent of bleeding and the patient's clinical condition.

The (functional) fibrinogen level should be determined in order to calculate individual dosage and the amount and frequency of administration should be determined on an individual patient basis by regular measurement of plasma fibrinogen level and continuous monitoring of the clinical condition of the patient and other replacement therapies used.

Normal plasma fibrinogen level is in the range of 1.5 – 4.5 g/l. The critical plasma fibrinogen level below which haemorrhages may occur is approximately 0.5 – 1.0 g/l. In case of major surgical intervention, precise monitoring of replacement therapy by coagulation assays is essential.

1. <u>Prophylaxis in patients with congenital hypo-, dys- or afibrinogenaemia and known bleeding tendency.</u>

   To prevent excessive bleeding during surgical procedures, prophylactic treatment is recommended to raise fibrinogen levels to 1 g/l and maintain fibrinogen at this level until haemostasis is secure and above 0.5 g/l until wound healing is complete.

   In case of surgical procedures or treatment of a bleeding episode, the dose should be calculated as follows:

   $$\text{Dose of fibrinogen (mg/kg body weight)} = \frac{[\text{Target level (g/l) - measured level (g/l)}]}{0.017 \text{ (g/l per mg/kg body weight)}}$$

   Subsequent posology (doses and frequency of injections) should be adapted based on the patient's clinical status and laboratory results.

   The biological half-life of fibrinogen is 3-4 days. Thus, in the absence of consumption, repeated treatment with human fibrinogen is not usually required. Given the accumulation that occurs in case of repeated administration for a prophylactis use, the dose and the frequency should be determined according to the therapeutic goals of the physician for a given patient.

Treatment of bleeding

Adults
For perioperative bleeding generally 2 g (or 30 mg/kg body weight) is administered, with subsequent infusions as required. In case of severe haemorrhage i.e. obstetric use / abruption placenta, large amounts (4 – 8 g) of fibrinogen may be required.

Children
The dosage should be determined according to the body weight and clinical need but is usually 20-30 mg/kg.

Method of Administration
Intravenous infusion or injection.
Haemocomplettan should be reconstituted according to section 6.6. The reconstituted solution should be warmed to room or body temperature before administration, then injected or infuseed slowly at a rate which the patient finds comfortable. The injection or infusion rate should not exceed approx. 5 ml per minute.

4.3 Contraindications

Hypersensitivity to the active substances or to any of the excipients listed in section 6.1.

Manifest thrombosis or myocardial infarction, except in cases of life-threatening haemorrhages.

4.4 Special warnings and special precautions for use

There is a risk of thrombosis when patients with congenital deficiency are treated with human fibrinogen, particularly with high dose or repeated dosing. Patients given human fibrinogen should be observed closely for signs or symptoms of thrombosis.

In patients with a history of coronary heart disease or myocardial infarction, in patients with liver disease, in peri- or post-operative patients, in neonates, or in patients at risk of thromboembolic events or disseminated intravascular coagulation, the potential benefit of treatment with human plasma fibrinogen should be weighed against the risk of thromboembolic complications. Caution and close monitoring should also be performed.

Acquired hypofibrinogenemia is associated with low plasma concentrations of all coagulation factors (not only fibrinogen) and inhibitors and so treatment with blood products containing coagulation factors should be considered (with or without administration of fibrinogen concentrate). Careful monitoring of the coagulation system is necessary.

If allergic or anaphylactic-type reactions occur, the injection/infusion should be stopped immediately. In case of anaphylactic shock, standard medical treatment for shock should be implemented.

In the case of replacement therapy with coagulation factors in other congenital deficiencies, antibody reactions have been observed, but there is currently no data with fibrinogen.

Important information about specific excipients of Haemocomplettan
Haemocomplettan contains up to 164 mg (7.1 mmol) sodium per 1g fibrinogen. This correlates with 11.5 mg (0.5 mmol) sodium per kg body weight of the patient if the recommended initial dose of 70 mg/kg body weight is applied. To be taken into consideration by patients on a controlled sodium diet.

Virus safety
Standard measures to prevent infections resulting from the use of medicinal products prepared from human blood or plasma include selection of donors, screening of individual donations and plasma pools for specific markers of infection and the inclusion of effective manufacturing steps for the inactivation/removal of viruses. Despite this, when medicinal products prepared from human blood or plasma are administered, the possibility of transmitting infective agents cannot be totally excluded. This also applies to unknown or emerging viruses and other pathogens.

The measures taken are considered effective for enveloped viruses such as human immunodeficiency virus (HIV), hepatitis B virus (HBV) and hepatitis C virus (HCV) and for the non-enveloped hepatitis A virus (HAV).

The measures taken may be of limited value against non-enveloped viruses such as parvovirus B19.
Parvovirus B19 infection may be serious for pregnant women (fetal infection) and for individuals with immunodeficiency or increased erythropoesis (e.g. haemolytic anaemia).

Appropriate vaccination (hepatitis A and hepatitis B) should be considered for patients in regular/repeated receipt of human fibrinogen products.

It is strongly recommended that every time that Haemocomplettan is administered to a patient, the name and batch number of the product are recorded in order to maintain a link between the patient and the batch of the product.

4.5 Interaction with other medicinal products and other forms of interaction

No interactions of human plasma fibrinogen products with other medicinal products are known.

4.6 Fertility, pregnancy and lactation

Pregnancy

Animal reproduction studies have not been conducted with Haemocomplettan (see section 5.3). Since the active substance is of human origin, it is catabolized in the same manner as the patient's own protein. These physiological constituents of the human blood are not expected to induce adverse effects on reproduction or on the fetus.

The safety of human plasma fibrinogen products for use in human pregnancy has not been established in controlled clinical trials.

Clinical experience with fibrinogen products in the treatment of obstetric complications suggests that no harmful effects on the course of the pregnancy or health of the fetus or the neonate are to be expected.

Lactation

It is unknown whether Haemocomplettan is excreted in human milk. The safety of human plasma fibrinogen products for use during lactation has not been established in controlled clinical trials.

A risk to the suckling child cannot be excluded. A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from Haemocomplettan therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

Fertility

There are no data regarding effects of Haemocomplettan on fertility.

4.7 Effects on ability to drive and use machines

Haemocomplettan has no influence on the ability to drive and use machines.

4.8 Undesirable effects

a. Summary of the safety profile

Allergic or anaphylactic type reactions have been uncommonly observed. The events reported in association with allergic/anaphylactic reactions include generalized urticarial, rash, dyspnoea, tachycardia, nausea, vomiting, chills, pyrexia, chest pain, cough, blood pressure decreased, and anaphylactic shock (see section 4.4).

The risk of thromboembolic events (TEE) following the administration of fibrinogen concentrate (see section 4.4.) as determined in clinical trial is further described in the table below.

Pyrexia has been very commonly observed.

b. Tabulated list of adverse drug reactions (ADRs)

This table combines the adverse reactions identified from clinical trials and post marketing experience. Frequencies presented in the table have been based on pooled analyses across two company sponsored clinical trials performed in aortic surgery with or without other surgical procedures (BI3023-2002 and BI3023_3002) according to the following convention: very common (≥1/10); common (≥1/100 to <1/10); uncommon (≥1/1,000 to <1/100); rare (≥1/10,000 to <1/1,000); very rare (<1/10,000) or unknown (cannot be estimated from the available data).

The calculated frequency is based on crude incidence rates without considering the frequency of the comparator arm. It should be noted that in the clinical trials included in the analysis, the incidence of TEEs was higher in the placebo arm. In view of the fact that these trials were conducted in only the narrow population of aortic surgery with or without other surgical procedures, adverse drug reaction rates observed in these trials may not reflect the rates observed in clinical practice and are unknown for clinical settings outside the studied indication.

MedDRA SOC	**Undesirable effects**	**Frequency** **(In aortic surgery with or without other surgical procedures)**
General Disorders and Administration Site Condition	Pyrexia	Very common
Immune System Disorder	Allergic or anaphylactic reactions	Uncommon
Vascular Disorder	Thromboembolic events*	Common**

** Isolated cases have been fatal.*
*** Based on results of two clinical trials (aortic surgery with or without other surgical procedures), the pooled incidence rate of thromboembolic events was lower in fibrinogen treated subjects compared with placebo (see 4.8.c).*

c. Description of selected adverse reactions (ADRs)

BI3023_2002 study is a Phase II study of fibrinogen concentrate human (FCH) compared with placebo (saline) in subjects with acute bleeding while undergoing aortic repair surgery. BI3023-3002 study is a Phase III study of FCH versus placebo (saline) to control bleeding during complex cardiovascular surgery. In BI3023_2002 study (N=61), TEE occurred similarly in fibrinogen and placebo groups. In BI3023_3002 study (N=152), TEE occurred more frequently in the placebo group than in the FCH group.

Pooled incidence rate of listed ADRs from company sponsored clinical trials (BI3023_2002 and BI3023_3002)

ADRs	FCH (N=107)	Placebo (N=106)
Pyrexia	11 (10.4%)	5 (4.7%)
Thromboembolic events	8 (7.4%)	11 (10.4%)
Allergic or anaphylactic reaction	1 (0.9%)	0

For safety with respect to transmissible agents, see section 4.4

Reporting of suspected adverse reactions
Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions.

4.9 Overdose

In order to avoid overdosage, regular monitoring of the plasma level of fibrinogen during therapy is indicated (see 4.2).
In case of overdosage, the risk of development of thromboembolic complications is enhanced.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group**:** antihaemorrhagics**,** human fibrinogen,
ATC code: B02B B01

Human fibrinogen (coagulation factor I), in the presence of thrombin, activated coagulation factor XIII (F XIIIa) and calcium ions, is converted into a stable and elastic three-dimensional fibrin haemostatic clot.

The administration of human fibrinogen provides an increase in plasma fibrinogen level and can temporarily correct the coagulation defect of patients with fibrinogen deficiency.

The pivotal Phase II study evaluated the single-dose PK (see 5.2 Pharmacokinetic properties) and also provided efficacy data using the surrogate endpoint maximum clot firmness (MCF) and safety data.
For each subject, the MCF was determined before (baseline) and one hour after a single dose administration of 70 mg/kg bw of Haemocomplettan. Haemocomplettan was found to be effective in increasing clot firmness in patients with congenital fibrinogen deficiency (afibrinogenaemia) as measured by thromboelastometry. Haemostatic efficacy in acute bleeding episodes, and its correlation with MCF, are being verified in a postmarketing study.

5.2 Pharmacokinetic properties

Human plasma fibrinogen is a normal constituent of the human plasma and acts like endogenous fibrinogen. In plasma, the biological half-life of fibrinogen is 3 to 4 days. Regarding degradation Haemocomplettan behaves like the endogenous fibrinogen.

Haemocomplettan is administered intravenously and is immediately available in a plasma concentration corresponding to the dosage administered.

A pharmacokinetic study evaluated the single-dose pharmacokinetics before and after administration of human fibrinogen concentrate in subjects with congenital afibrinogenaemia. This prospective, open label, uncontrolled, multicenter study consisted of 5 females and 10 males, ranging in age from 8 to 61 years (2 children, 3 adolescents, 10 adults). The median dose was 77.0 mg/kg body weight (range 76.6 – 77.4 mg/kg).

Blood was sampled from 15 subjects (14 measurable) to determine the fibrinogen activity at baseline and up to 14 days after the infusion was complete. In addition, the incremental *in vivo* recovery (IVR), defined as the maximum increase in fibrinogen plasma levels per mg/kg body weight dosed, was determined from levels obtained up to 4 hours post-infusion. The median incremental IVR was 1.7 (range 1.30-2.73) mg/dl per mg/kg body weight. The following table provides the pharmacokinetic results.

Pharmacokinetic results for fibrinogen activity

Parameter (n=14)	Mean ± SD	Median (range)
$t_{1/2}$ [h]	78.7 ± 18.13	77.1 (55.73-117.26)
C_{max} [g/l]	1.4 ± 0.27	1.3 (1.00-2.10)
AUC for dose of 70 mg/kg [h•mg/ml]	124.3 ± 24.16	126.8 (81.73-156.40)
Extrapolated part of AUC [%]	8.4 ± 1.72	7.8 (6.13-12.14)
Cl [ml/h/kg]	0.59 ± 0.13	0.55 (0.45-0.86)
MRT [h]	92.8 ± 20.11	85.9 (66.14-126.44)
V_{ss} [ml/kg]	52.7 ± 7.48	52.7 (36.22-67.67)
IVR [mg/dl per mg/kg body weight]	1.8 ± 0.35	1.7 (1.30-2.73)

$t_{1/2}$ = terminal elimination half-life
h = hour
C_{max} = maximum concentration within 4 hours
AUC = area under the curve
Cl = clearance
MRT = mean residence time
V_{ss} = volume of distribution at steady state
SD = standard deviation
IVR = in vivo recovery

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of single dose toxicity and safety pharmacology.

Preclinical studies with repeated dose applications (chronic toxicity, cancerogenicity and mutagenicity) cannot be reasonably performed in conventional animal models due to the development of antibodies following the application of heterologous human proteins.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Human albumin,
L-arginine hydrochloride,
sodium hydroxide (for pH adjustment),
sodium chloride,
sodium citrate.

6.2 Incompatibilities

This medicinal product must not be mixed with other medicinal products, diluents, or solvents except those mentioned in section 6.6. A standard infusion set is recommended for intravenous application of the reconstituted solution at room temperature.

6.3 Shelf life

5 years.

The physico-chemical stability for the reconstituted product has been demonstrated for 8 hours at room temperature (max. +25 °C). From a microbiological point of view the product should be used immediately following reconstitution. If the reconstituted product is not administered immediately, storage shall not exceed 8 hours at room temperature (max. +25 °C). The reconstituted product should not be stored in the refrigerator.

6.4 Special precautions for storage

Store in a refrigerator (2 °C – 8 °C). Do not freeze! Keep the vial in the outer carton, in order to protect from light.

6.5 Nature and contents of container

Vials of colourless glass (Type II Ph. Eur.) sealed with a latex-free stopper (bromobutyl rubber), aluminium cap and plastic disc.

Pack with 1 or 2 g *(Figure 1)*

1. One vial containing 1 or 2 g human fibrinogen
2. Filter: Pall® Syringe Filter
3. Dispensing pin: Mini-Spike® Dispensing Pin

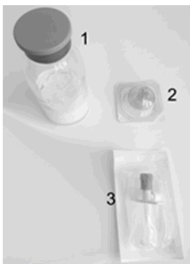


Figure 1

6.6 Instructions for use, handling and disposal

General instructions

- Reconstitution and withdrawal should be carried out under aseptic conditions.
- Reconstituted products should be inspected visually for particulate matter and discoloration prior to administration.
- The solution should be almost colourless to yellowish, clear to slightly opalescent and of neutral pH. Do not use solutions that are cloudy or have deposits.

Reconstitution

- Warm both the solvent and the powder in unopened vials to room or body temperature (not above 37 °C).
- Haemocomplettan should be reconstituted with water for injections (50 ml for 1 g and 100 ml for 2 g, respectively, not included).
- Wash hands or use gloves before reconstituting the product.
- Remove the cap from the Haemocomplettan vial to expose the central portions of the infusion stoppers.
- Treat the surface of the infusion stopper with antiseptic solution and allow it to dry.
- Transfer the solvent with an appropriate transfer device into the infusion vial. Ensure complete wetting of the powder.
- Gently swirl the vial until the powder is reconstituted and the solution is ready for administration. Avoid strong shaking which causes formation of foam. The powder should be completely reconstituted within max. 15 minutes (generally within 5 to 10 minutes).
- Open the plastic blister containing the dispensing pin (Mini-Spike® Dispensing Pin) provided with Haemocomplettan (Figure 2).


Figure 2

- Take the provided dispensing pin and insert it into the stopper of the vial with the reconstituted product (Figure 3).


Figure 3

- After the dispensing pin is inserted, remove the cap. After the cap is removed, do not touch the exposed surface.
- Open the blister with the filter (Pall® Syringe Filter) provided with Haemocomplettan (Figure 4).

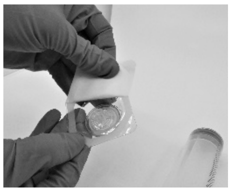


Figure 4

- Screw the syringe onto the filter (Figure 5).


Figure 5

- Screw the syringe with the mounted filter onto the dispensing pin (Figure 6).


Figure 6

- Draw the reconstituted product into the syringe (Figure 7).


Figure 7

- When completed, **remove the filter, dispensing pin and empty vial from the syringe,** dispose of properly, and proceed with administration as usual.
- Reconstituted product should be administered immediately by a separate injection / infusion line (see section 6.3).

- Take care that no blood enters syringes filled with product.

Any unused product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

CSL Behring GmbH
Emil-von-Behring-Str. 76
35041 Marburg
Germany

8. DATE OF (PARTIAL) REVISION OF THE TEXT

February 2021

9. **Imported And Marketed In India By: Plasmagen Biosciences Private Limited**, No. 1342, Block B, 1st Floor, Dr. Shivaramkaranth Nagar, M.C.E.H.S Layout, Yelahanka Hobli, Bangalore - 560064, Karnataka, India